Unit 6, Lesson 8

Multiply 2 Two-Digit Numbers

Standards

Addressing	**4.NBT.B.5**
Building Toward	**4.NBT.B.5**

Instructional Routines

- MLR5 Co-Craft Questions
- Number Talk

Goals

- Describe (orally) how decomposing factors by place value is helpful when determining products.
- Use a diagram to multiply 2 two-digit numbers.

Student Facing Learning Goals

Let's multiply 2 two-digit numbers.

Lesson Purpose

The purpose of this lesson is for students to multiply 2 two-digit numbers.

Narrative

Previously, students used place-value reasoning to decompose a factor in a multiplication expression to multiply numbers up to four-digit by one-digit numbers. In this lesson, they apply these ideas to multiply 2 two-digit numbers. They reason about why it is helpful to decompose both two-digit numbers by place value. As students analyze the connections between expressions and diagrams, they recognize that partial products in which the factors are either single-digit numbers or multiples of 10 can be found mentally, making the rectangular diagram a useful tool for multiplying two-digit numbers.

Access for Students with Disabilities

- Action and Expression

Access for English Learners

- MLR8

Lesson Timeline

Warm-up	10 min
Activity 1	20 min
Activity 2	15 min
Synthesis Estimate	10 min
Cool-down	5 min

Teacher Reflection Questions

In a future lesson, students will be analyzing partial products from rectangular diagrams and making connections to the traditional algorithm notation. How do rectangular diagrams support this thinking?

Warm-up

10 min

Number Talk: Extra Groups

Standards

Building Toward **4.NBT.B.5**

Instructional Routines

- Number Talk

This *Number Talk* encourages students to use multiples of 10 to mentally multiply two-digit numbers that are close to multiples of 10. Students can use place value reasoning and the distributive property of multiplication over addition or subtraction to find the value of the products. Students think flexibly about how numbers can be decomposed strategically when multiplying.

Student Task Statement

Find the value of each expression mentally.

- 20×60
- 21×60
- 20×62
- 19×60

Launch

- Display one expression.
- "Give me a signal when you have an answer and can explain how you got it."
- 1 minute: quiet think time

Activity

- Record answers and strategy.
- Keep expressions and work displayed.
- Repeat with each expression.

Student Response

- 1,200: I know 2×6 is 12, so $2 \times 10 \times 6 \times 10$ is 12×100, which is 1,200.
- 1,260: 21×60 is one more group of 60 than 20×60, so it is $1{,}200 + 60$ or 1,260 (or $(20 \times 60) + (1 \times 60) = 1{,}260$).
- 1,240: 20×62 is 20×2 more than 20×60, so it is 40 more than 1,200 (or $(20 \times 60) + (20 \times 2) = 1{,}240$).
- 1,140: 19 groups of 60 is 1 group of 60 less than 20 groups of 60, so it is $1{,}200 - 60$ or 1,140 (or $(20 \times 60) - (1 \times 60) = 1{,}140$).

Activity Synthesis

- "How can 20×60 help us find the value of 19×60?" (19×60 is one group of 60 less than 20×60, so we can subtract 60 from 20×60.)

Activity 1

20 min

Two by Two

Standards

Addressing **4.NBT.B.5**

In this activity, students use rectangular diagrams and similar reasoning as in earlier activities to represent the multiplication of 2 two-digit numbers. They analyze a progression of diagrams, starting with those that represent multiplication of two-digit and one-digit numbers (18 and 6), a two-digit number and a ten (18 and 10), and then 2 two-

digit numbers (18 and 16).

Students may decompose factors in different ways. For example, those who are familiar with multiples of 25 may find it intuitive to decompose 25×46 as 25×40 and 25×6, rather than decomposing 25 into $20 + 5$.

Access for English Language Learners

MLR8 Discussion Supports. Pair gestures with verbal directions to clarify the meaning of any unfamiliar terms such as partial product.
Advances: Listening, Representing

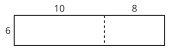
Access for Students with Disabilities

Action and Expression: Develop Expression and Communication. Give students access to base-ten blocks. If students use the blocks for the last question, ask them to also draw a diagram that represents their work with the blocks.
Supports accessibility for: Conceptual Processing, Visual-Spatial Processing

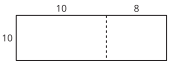
Student Task Statement

1. For each diagram, write a multiplication expression that the diagram can represent. Then find the value of the expression. Use equations to explain or show your reasoning.

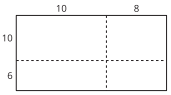
   a.

   b.

   c.

2. How are these diagrams alike? How are they different? Discuss with your partner.
3. Use a diagram to find each product.
   - 13×21
   - 25×46

Student Response

1. a. 18×6 or 6×18. The value is 108. Sample response: $(10 \times 6) + (8 \times 6) = 60 + 48 = 108$

Launch

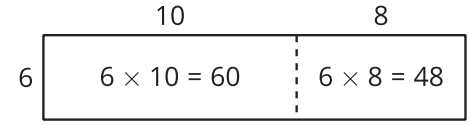
- Display the three diagrams in the activity.
- "What do you notice? What do you wonder?"
- 1 minute: quiet think time
- 1 minute: partner discussion

Activity

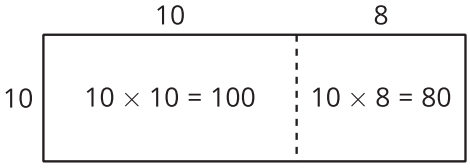
- "Take a few quiet minutes to work on the first problem. Then share your thinking with your partner."
- 3 minutes: independent work time
- 3 minutes: partner discussion
- Invite students to share their responses: "What multiplication expression can be represented by each diagram?"
- "Complete the rest of the activity."
- 3 minutes: independent or partner work time
- Monitor for students who decompose both factors into tens and ones and those who choose to keep one of the factors intact.

Activity Synthesis

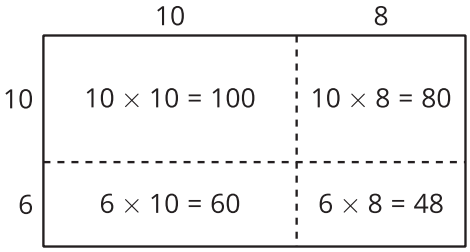
- Select two students to share their diagrams, solutions, and reasoning for the last problem.
- "How did you decompose the factors and the



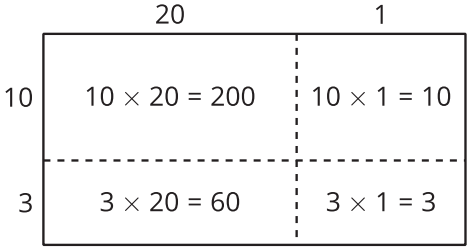
b. 18×10 or 10×18. The value is 180. Sample response:
$(10 \times 10) + (10 \times 8) = 100 + 80 = 180$



c. 18×16 or 16×18. The value is 288. Sample response:
$(10 \times 10) + (10 \times 8) + (6 \times 10) + (6 \times 8) = 100 + 80 + 60 + 48 = 288$



2. Sample response: The diagrams are alike because they all show multiplication by 18, decomposed by place value. They are different because Diagram A shows a one-digit factor, B shows 10 as a factor, and C shows a two-digit factor partitioned into two rows. Diagrams A and B are smaller parts of Diagram C.
3. ◦ 13×21:

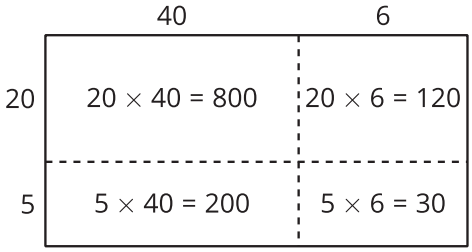


$200 + 10 + 60 + 3 = 273$

◦ 25×46:

diagram?" (I decomposed the 13 into 10 and 3 and the 21 into 20 and 1.)

- "What expression could we write to show the partial product represented by each part of the diagram?"
- "How do these partial products help us find the value of 13×21?" (Adding them gives us the value of 13×21: $200 + 10 + 3 + 60 = 273$)



$800 + 120 + 200 + 30 = 1{,}150$

Activity 2

15 min

Decompose by Place Value

Standards

Addressing **4.NBT.B.5**

Instructional Routines

- MLR5 Co-Craft Questions

In this activity, students analyze two ways of decomposing a factor: by place value and not by place value. As they write the corresponding partial products, they see more clearly why it is helpful to decompose each of the factors by place value (MP7). Students may notice that when the factors are decomposed by place value, they end up finding multiples of 10 and multiplying a number by single-digit factors—both of which they can do with some degree of fluency.

This activity uses *MLR5 Co-craft Questions.* Advances: writing, reading, representing

Student Task Statement

These diagrams could be used to find the value of 49×57.

Diagram A

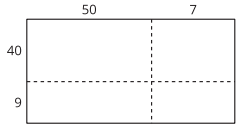
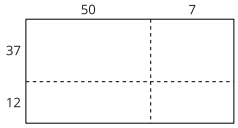


Diagram B



1. Which diagram is more helpful when finding the value of 49×57? Why?
2. Use a diagram to find each product.
   - 49×57
   - 29×55

Student Response

1. Sample response: Diagram A, because 40×50 is easier than 37×50 to find or know.
2. - $2{,}000 + 280 + 450 + 63 = 2{,}793$

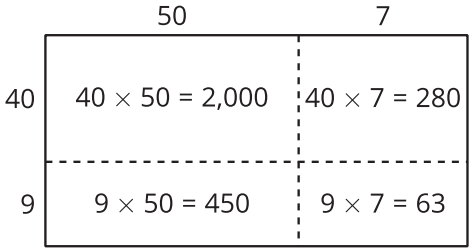
Launch

- Groups of 2

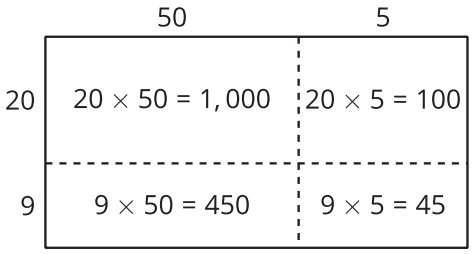
Activity

MLR5 Co-Craft Questions

- Display only the two diagrams, without revealing the opening sentence or the question(s).
- "Write a list of mathematical questions that could be asked about this situation."
- 2 minutes: independent work time
- 2 minutes: partner discussion
- Invite several students to share one question with the class. Record responses.
- "What do these questions have in common? How are they different?"
- Reveal the task (students open books), and invite additional connections.
- "Take two quiet minutes to think about the first question. Then share your thinking with your



- $1{,}000 + 100 + 450 + 45 = 1{,}595$



partner."

- 2 minutes: independent work time on the first problem
- 1 minute: partner discussion
- "Now complete the rest of the activity."
- Monitor for students who:
  - Decompose the side lengths of their diagrams by place value.
  - Write expressions to show the sum of the partial products.
- If extra time is available, add more two-digit by two-digit expressions to the last problem:
  - 83×39
  - 64×92

Activity Synthesis

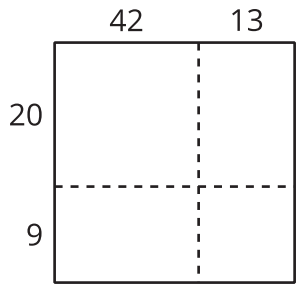
- "How can both diagrams be used to find the value of 49×57?" (We can partition each side of a rectangle in many ways without changing the total side lengths.)
- "Why was the partitioning in Diagram A more helpful than in Diagram B?" (40×50 is easier than 37×50 or 12×50 to find mentally, because we are multiplying multiples of 10.)

Lesson Synthesis

"Today we learned how to represent the multiplication of 2 two-digit numbers using a rectangular diagram. We learned that we can decompose each factor by place value and show the tens and ones on each side of the rectangle, and that doing this can help us to multiply efficiently."

Select students with different strategies to share their reasoning for finding the value of 29×55 (the last problem of the last activity).

Display the following diagram as an example of how decomposing can result in facts that are not helpful when multiplying to support using place value to decompose.



"Why might it be more helpful to decompose 55 into $50 + 5$ than into $42 + 13$?" (Multiplying by multiples of 10 and by

single-digit numbers is easier than multiplying numbers like 42 and 9.)

Suggested Centers

- Number Puzzles: Multiplication (4–5) , Stage 1: Two-Digit Factors (Addressing)
- Compare (1–5), Stage 3: Multiply within 100 (Supporting)

Cool-down

5 min

What's the Product?

Standards

Addressing **4.NBT.B.5**

Student Task Statement

Find the value of 24×17. Explain or show your reasoning. Use a diagram if it is helpful.

Student Response

408. Sample response:

$200 + 140 + 40 + 28 = 408$

Responding to Student Thinking

Students may decompose factors in ways to result in partial products that are not useful.

Next Day Supports
Launch *Warm-up* or activities by analyzing two different ways to decompose the factors to multiply. Discuss which is most useful and why.